INCH POUND

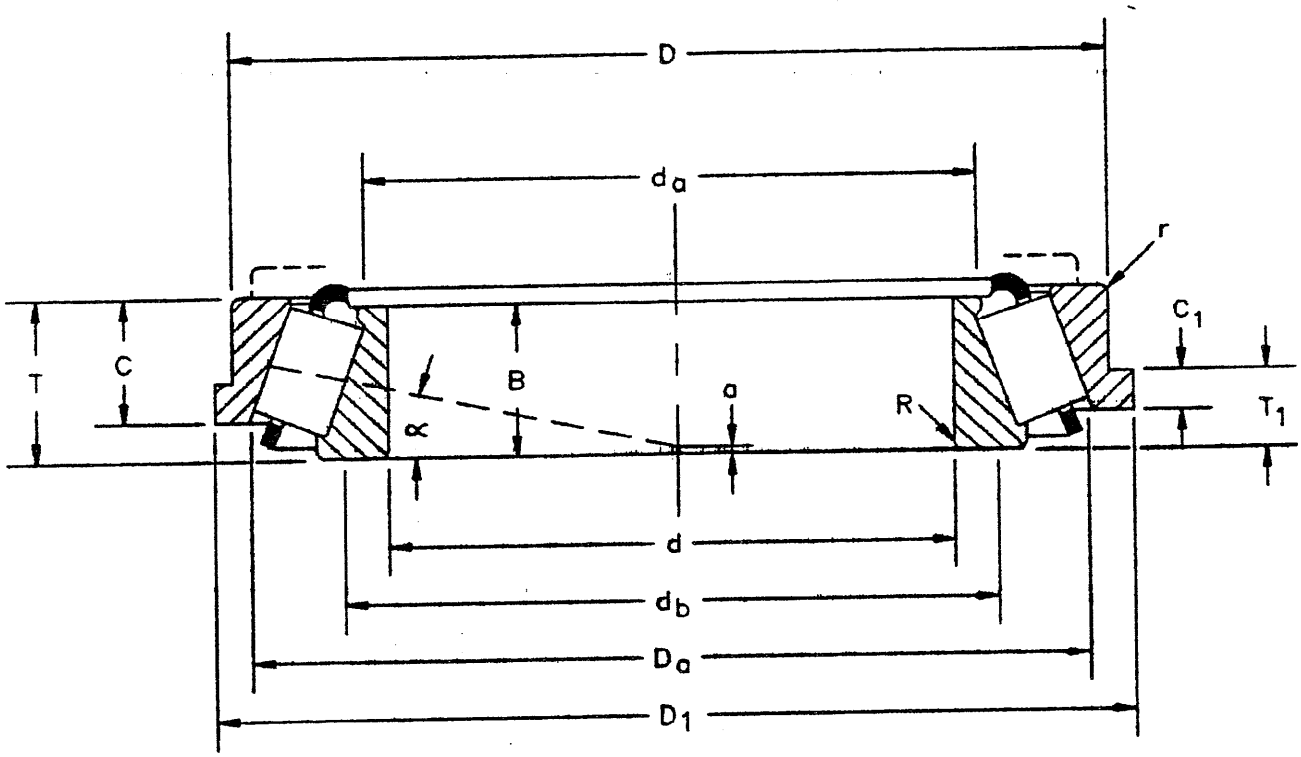
FF-B-187/2
10 SEPTEMBER 1993

FEDERAL SPECIFICATION SHEET

BEARING, ROLLER, TAPERED, SINGLE ROW, NORMAL ANGLE, FLANGED CUP (TYPE TSF)

This specification is approved by the Administrator, Federal Supply Service, General Services Administration, for the use of all federal Agencies.

The complete requirements for procuring the bearings described herein, shall consist. of this document and the latest issue of FF-B-187/GEN.



BEARING CONFIGURATION

FSC 3110

AMSC N/A

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TABLE I. DASH NUMBERS AND DIMENSIONS

DASH NUMBER	PART NUMBER 1/ CONE-CUP	d BORE	D OUTSIDE DIA	T BEARING WIDTH	T1 STANDOUT	B CONE WIDTH	C CUP WIDTH	R 2/ MAX SHAFT FILLET RADIUS	D1 FLANGE DIA	C1 FLANGE WIDTH	db BACKING SHOULDER DIAMETERS SHAFT	da BACKING SHOULDER DIAMETERS SHAFT	Da BACKING SHOULDER DIAMETERS HOUSING	K FACTOR	BASIC DYNAMIC LOAD RATINGS (LB) RADIAL	a 3/ EFFECTIVE LOAD CENTER
001	A4050-A4138-B	0.5000	1.3775	0.4330	0.1823	0.4326	0.3437	0.05	1.4985	0.0930	0.73	0.67	1.32	1.29	2100	-0.10
002	A4059-A4138-B	0.5906	1.3775	0.4330	0.1823	0.4326	0.3437	0.03	1.4985	0.0930	0.77	0.75	1.32	1.29	2100	-0.10
003	17580-17520-B	0.6250	1.6875	0.6563	0.2500	0.6563	0.5315	0.06	1.8085	0.1250	0.91	0.83	1.59	1.76	5100	-0.23
004	05062-05185-B	0.6250	1.8504	0.5662	0.2377	0.5662	0.4375	0.06	2.0024	0.1090	0.93	0.83	1.75	1.84	4400	-0.16
005	A6075-A6157-B	0.7500	1.5745	0.4730	0.1910	0.4391	0.3750	0.04	1.6955	0.0930	0.94	0.91	1.50	1.11	2200	-0.06
006	09067-09195-A-B	0.7500	1.9380	0.7100	0.2755	0.7500	0.5625	0.05	2.0898	0.1280	1.00	0.94	1.83	2.20	6900	-0.29
007	1380-1328-B	0.8750	2.0625	0.7625	0.3560	0.7940	0.5625	0.06	2.2147	0.1560	1.16	1.06	1.97	2.00	7800	-0.30
008	1779-1729-B	0.9375	2.2400	0.7625	0.2935	0.7810	0.6250	0.03	2.3920	0.1560	1.16	1.12	2.09	1.90	7300	-0.27
009	3659-3620-B	0.9375	2.4375	1.1250	0.3750	1.1875	0.9375	0.09	2.6209	0.1875	1.40	1.24	2.32	2.07	14200	-0.47
010	07100-07196-B	1.0000	1.9687	0.5313	0.2657	0.5614	0.3750	0.04	2.1209	0.1094	1.20	1.16	1.93	1.45	4800	-0.11
011	07100-07204-B	1.0000	2.0470	0.5910	0.2000	0.5614	0.5000	0.04	2.1990	0.1090	1.20	1.16	1.97	1.45	4800	-0.11
012	1718-1729-B	1.0000	2.2400	0.7625	0.2935	0.7810	0.6250	0.03	2.3920	0.1560	1.20	1.18	2.09	1.90	7300	-0.27
013	15101-15250-B	1.0000	2.5000	0.8125	0.3437	0.8125	0.6250	0.03	2.6522	0.1562	1.28	1.24	2.36	1.67	8200	-0.23
014	15580-15520-B	1.0625	2.2500	0.6875	0.3125	0.6875	0.5313	0.14	2.4022	0.1563	1.52	1.26	2.17	1.69	7500	-0.20
015	15590-15520-B	1.1250	2.2500	0.6875	0.3125	0.6875	0.5313	0.14	2.4022	0.1563	1.56	1.32	2.17	1.69	7500	-0.20
016	1985-1931-B	1.1250	2.3750	0.7813	0.3125	0.7620	0.6250	0.03	2.5270	0.1562	1.34	1.32	2.24	1.77	8300	-0.23
017	02474-02420-B	1.1250	2.6875	0.8750	0.3437	0.8750	0.6875	0.03	2.8397	0.1562	1.44	1.42	2.56	1.40	10400	-0.20
018	3198-3120-B	1.1250	2.8593	1.1875	0.4375	1.1810	0.9375	0.05	3.0433	0.1875	1.54	1.46	2.72	1.76	15200	-0.40
019	17119-17244-B	1.1875	2.4409	0.6300	0.2075	0.6522	0.5625	0.06	2.5930	0.1400	1.46	1.36	2.32	1.53	7000	-0.14
020	02476-02420-B	1.2500	2.6875	0.8750	0.3437	0.8750	0.6875	0.03	2.8397	0.1562	1.54	1.52	2.32	1.40	10400	-0.20
021	02475-02420-B	1.2500	2.6875	0.8750	0.3437	0.8750	0.6875	0.14	2.8397	0.1562	1.75	1.52	2.56	1.40	10400	-0.20
022	2580-2523-B	1.2500	2.7500	0.9375	0.3435	0.9983	0.7500	0.03	2.9020	0.1560	1.52	1.48	2.60	2.14	13500	-0.34

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2/ THESE MAXIMUM FILLET RADII SHALL BE CLEARED BY THE BEARING CORNERS.

3/ MINUS VALUE INDICATES LOAD CENTER INSIDE CONE BACKFACE.

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TABLE I. DASH NUMBERS AND DIMENSIONS - Continued

DASH NUMBER	PART NUMBER 1/ CONE-CUP	d BORE	D OUTSIDE DIA	T BEARING WIDTH	T1 STANDOUT	B CONE WIDTH	C CUP WIDTH	R 2/ MAX SHAFT FILLET RADIUS	D1 FLANGE DIA	C1 FLANGE WIDTH	db BACKING SHOULDER DIAMETERS SHAFT	da BACKING SHOULDER DIAMETERS SHAFT	Da BACKING SHOULDER DIAMETERS HOUSING	K FACTOR	BASIC DYNAMIC LOAD RATINGS (LB) RADIAL	a 3/ EFFECTIVE LOAD CENTER
023	2581-2523-B	1.3125	2.7500	0.9375	0.3435	0.9983	0.7500	0.03	2.9020	0.1560	1.56	1.54	2.60	2.14	13500	-0.34
024	2585-2523-B	1.3125	2.7500	0.9375	0.3435	0.9983	0.7500	0.14	2.9020	0.1560	1.77	1.54	2.60	2.14	13500	-0.34
025	3196-3120-B	1.3125	2.8593	1.1875	0.4375	1.1810	0.9375	0.14	3.0433	0.1875	1.85	1.59	2.72	1.76	15200	-0.40
026	31594-31520-B	1.375	3.0000	1.1563	0.4063	1.1250	0.9375	0.06	3.1836	0.1875	1.181	1.71	2.91	1.45	15200	-0.30
027	31597-31520-B	1.4375	3.0000	1.1563	0.4063	1.1250	0.9375	0.14	3.1836	0.1875	2.01	1.75	2.91	1.45	15200	-0.30
028	13889-13836-B	1.5000	2.5625	0.5000	0.2340	0.4688	0.3750	0.06	2.6835	0.1090	1.77	1.67	2.48	1.69	4500	-0.03
029	19150-19268-B	1.5000	2.6875	0.6250	0.2968	0.6504	0.4688	0.06	2.8397	0.1406	1.77	1.69	2.64	1.31	8300	-0.06
030	19150-19283-B	1.5000	2.8346	0.6700	0.2475	0.6504	0.5625	0.06	2.9865	0.1400	1.77	1.69	2.68	1.31	8300	-0.06
031	16150-16484-B	1.5000	2.8440	0.8125	0.3438	0.8125	0.6250	0.14	2.9960	0.1563	1.95	1.69	2.72	1.45	9200	-0.16
032	3490-3420-B	1.5000	3.1250	1.1563	0.4063	1.1721	0.9375	0.14	3.3090	0.1875	2.05	1.80	2.99	1.60	16800	-0.34
033	28150-28315-B	1.5000	3.1496	0.8270	0.0583	0.8244	0.6250	0.06	3.3015	0.1563	1.79	1.71	2.87	1.45	16000	-0.19
034	3382-3320-B	1.5625	3.1562	1.1563	0.4063	1.1965	0.9375	0.14	3.3396	0.1875	2.05	1.79	3.03	2.14	18500	-0.43
035	11162-11300-B	1.6250	3.0000	0.7090	0.2871	0.6844	0.5625	0.06	3.1836	0.1406	1.93	1.83	2.87	1.20	7800	-0.03
036	26882-26822-B	1.6250	3.1250	0.9375	0.3438	0.1000	0.7500	0.14	3.2772	0.1563	2.13	1.85	2.99	1.83	14700	-0.29
037	3877-3820-B	1.6250	3.3750	1.1875	0.4375	1.1875	0.9375	0.14	3.5270	0.1875	2.24	1.98	3.27	1.45	20000	-0.32
038	LL103049-LL103010-B	1.7500	2.8125	0.5000	0.2350	0.5000	0.3750	0.06	2.9225	0.1100	2.01	1.91	2.72	1.91	5900	-0.05
039	L102849-L102810-B	1.7500	2.8750	0.7188	0.2500	0.7188	0.5938	0.06	3.0000	0.1250	2.01	1.93	2.80	1.83	9300	-0.15
040	3578-3525-B	1.7500	3.4375	1.1875	0.4370	1.2160	0.9375	0.14	3.6215	0.1870	2.24	2.01	3.23	1.91	18100	-0.40

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TABLE I. DASH NUMBERS AND DIMENSIONS -Continued

DASH NUMBER	PART NUMBER 1/ CONE-CUP	d BORE	D OUTSIDE DIA	T BEARING WIDTH	T1 STANDOUT	B CONE WIDTH	C CUP WIDTH	R 2/ MAX SHAFT FILLET RADIUS	D1 FLANGE DIA	C1 FLANGE WIDTH	db BACKING SHOULDER DIAMETERS SHAFT	da BACKING SHOULDER DIAMETERS SHAFT	Da BACKING SHOULDER DIAMETERS HOUSING	K FACTOR	BASIC DYNAMIC LOAD RATINGS (LB) RADIAL	a 3/ EFFECTIVE LOAD CENTER
041 •	49175-49368-B	1.7500	3.6875	1.2500	4.6885	1.2500	1.0000	0.14	3.9073	0.2188	2.32	2.09	3.50	1.62	22000	-0.36
042	438-432-B	1.7500	3.7500	1.0938	0.4376	1.1772	0.8750	0.14	3.9640	0.2188	2.24	2.01	3.43	2.05	20500	-0.36
043	18690-18620-B	1.8125	3.1250	0.6875	0.2968	0.6875	0.5313	0.11	3.3086	0.1406	2.20	2.01	3.03	1.56	8500	-0.08
044	25592-25521-B	1.8125	3.2700	0.9375	0.3438	1.0000	0.7500	0.14	3.4220	0.1563	2.28	2.05	3.15	1.74	14500	-0.25
045	359-S-354-B	1.8125	3.3465	0.8125	0.3125	0.8540	0.6875	0.09	3.5299	0.1875	2.17	2.01	3.23	1.91	13300	-0.19
046	2984-2924-B	1.8125	3.3465	1.0000	0.3750	1.0082	0.8125	0.14	3.5340	0.1875	2.28	2.05	3.23	1.69	15100	-0.25
047	436-432-B	1.8125	3.7500	1.0938	0.4376	1.1772	0.8750	0.14	3.9640	0.2188	2.32	2.05	3.43	2.05	20500	-0.36
048	65390-65320-B	1.9375	4.5000	1.7500	0.6563	1.7500	1.3750	0.14	4.7772	0.2813	2.76	2.36	4.21	1.36	35700	-0.49
049	L305649-L305610-B	2.0000	3.1875	0.7188	0.2813	1.7188	0.5625	0.06	3.3085	0.1250	2.28	2.20	3.07	1.64	9900	-0.10
050	18790-18720-B	2.0000	3.3465	0.6875	0.2968	0.6875	0.5313	0.14	3.4870	0.1406	2.44	2.20	3.23	1.44	8500	-0.03
051	368A-362-B	2.0000	3.5433	0.7874	0.3499	0.8750	0.6250	0.14	3.7268	0.1875	2.44	2.20	3.39	1.83	14000	-0.17
052	3780-3720-B	2.0000	3.6718	1.1875	0.4375	1.1930	0.9375	0.14	3.8558	0.1875	2.52	2.28	3.54	1.73	19700	-0.32
053	49585-49520-B	2.0000	4.0000	1.2500	0.4688	1.2500	1.0000	0.14	4.2148	0.2188	2.60	2.32	3.86	1.46	21300	-0.28
054	529-522-B	2.0000	4.0000	1.3750	0.5625	1.4200	1.0625	0.03	4.2460	0.2500	2.32	2.28	3.82	2.05	26400	-0.50
055	529X-522-B	2.0000	4.0000	1.3750	0.5625	1.4200	1.0625	0.14	4.2460	0.2500	2.56	2.28	3.82	2.05	26400	-0.50
056	65395-65320-B	2.0000	4.5000	1.7500	0.6563	1.7500	1.3750	0.14	4.7772	0.2813	2.83	2.36	4.21	1.36	35700	-0.49
057	28584-28521-B	2.0625	3.6250	0.9688	0.3437	1.0000	0.7813	0.14	3.7772	0.1582	2.56	2.28	3.50	1.55	16000	-0.19
058	33895-33820-B	2.1250	3.6718	1.0938	0.4063	1.1250	0.8750	0.06	3.8558	0.1875	2.48	2.36	3.58	1.77	20900	-0.30
059	539-532-B	2.1250	4.3750	1.5000	0.5625	1.4550	1.1875	0.14	4.6210	0.2500	2.68	2.40	3.94	1.97	27600	-0.48
060	621-612-B	2.1250	4.7500	1.6250	0.6562	1.6250	1.2500	0.14	5.0272	0.2812	2.76	2.48	4.33	1.86	33200	-0.55

• INACTIVE FOR NEW DESIGN AFTER 20 MAR 1978.

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TABLE I. DASH NUMBERS AND DIMENSIONS - Continued

DASH NUMBER	PART NUMBER 1/ CONE-CUP	d BORE	D OUTSIDE DIA	T BEARING WIDTH	T1 STANDOUT	B CONE WIDTH	C CUP WIDTH	R 2/ MAX SHAFT FILLET RADIUS	D1 FLANGE DIA	C1 FLANGE WIDTH	db BACKING SHOULDER DIAMETERS SHAFT	da BACKING SHOULDER DIAMETERS SHAFT	Da BACKING SHOULDER DIAMETERS HOUSING	K FACTOR	BASIC DYNAMIC LOAD RATINGS (LB) RADIAL	a 3/ EFFECTIVE LOAD CENTER
061	L507949-L507910-B	2.2500	3.4375	0.7188	0.2813	0.7188	0.5625	0.06	3.5625	0.1250	2.56	2.44	3.35	1.50	9400	-0.03
062	387A-382-B	2.2500	3.8125	0.8268	0.3125	0.8640	0.7018	0.14	3.9960	0.1875	2.72	2.44	3.70	1.65	14800	-0.12
063	28682-28622-B	2.2500	3.8437	0.9688	0.3592	0.9688	0.7656	0.14	3.9960	0.1560	2.76	2.48	3.70	1.45	16800	-0.13
064	469-453-B	2.2500	4.2500	1.0938	0.4376	1.1542	0.8750	0.14	4.4640	0.2188	2.76	2.48	3.94	1.74	21800	-0.28
065	623-612-B	2.2500	4.7500	1.6250	0.6562	1.6250	1.2500	0.14	5.0272	0.2812	2.83	2.60	4.33	1.86	33200	-0.55
066	555S-552-B	2.2500	4.8750	1.5000	0.5625	1.4440	1.1875	0.14	5.1210	0.2500	2.87	2.64	4.57	1.69	30700	-0.37
067	28985-28921-B	2.3750	3.9370	1.0000	0.3750	1.0000	0.7813	0.14	4.0930	0.1563	2.87	2.64	3.86	1.37	17200	-0.10
068	65237-65500-B	2.3750	5.0000	1.7500	0.6563	1.7500	1.3750	0.14	5.2772	0.2813	3.23	2.79	4.72	1.20	33300	-0.37
069	39250-39412-B	2.5000	4.1250	0.8438	0.4063	0.8661	0.6250	0.08	4.3084	0.1875	2.87	2.72	4.02	1.51	15800	-0.06
070	29585-29520-B	2.5000	4.2500	1.0000	0.4063	1.0000	0.7500	0.14	4.4022	0.1563	3.03	2.80	4.13	1.27	14200	-0.03
071	390A-394A-B	2.5000	4.3307	0.8661	0.3120	0.8660	0.7411	0.06	4.5147	0.1870	2.87	2.76	4.17	1.45	16200	-0.03
072	29585-29521-B	2.5000	4.3307	1.0000	0.4063	1.0000	0.7500	0.14	4.4803	0.1563	3.03	2.80	4.13	1.27	14200	-0.03
073 *	3984-3920-B	2.5000	4.8125	1.1875	0.4375	1.1830	0.9375	0.14	4.6210	0.1875	3.03	2.80	4.25	1.45	22400	-0.18
074 *	5595-5535-B	2.5938	4.8125	1.7188	0.5313	1.7230	1.4375	0.14	5.0625	0.2500	3.27	3.03	4.65	1.27	14200	-0.48
075	29590-29520-B	2.6250	4.2500	1.0000	0.4063	1.0000	0.7500	0.14	4.4022	0.1563	3.15	2.87	4.13	1.27	14200	-0.03
076	395A-394A-B	2.6250	4.3307	0.8661	0.3120	0.8660	0.7411	0.03	4.5147	0.1870	2.87	2.87	4.17	1.45	16200	-0.03
077	395-S-394A-B	2.6250	4.3307	0.8661	0.3120	0.8660	0.7411	0.14	4.5147	0.1870	3.11	2.87	4.17	1.45	16200	-0.03
078	29590-29521-B	2.6250	4.3307	1.0000	0.4063	1.0000	0.7500	0.14	4.4803	0.1563	3.15	2.87	4.13	1.27	14200	-0.03
079	3984-3920-B	2.6250	4.4375	1.1875	0.4375	1.1830	0.9375	0.14	4.6210	0.1875	3.15	2.91	4.25	1.45	22400	-0.18
080	39590-39520-B	2.6250	4.4375	1.1875	0.4688	1.1875	0.9375	0.14	4.6523	0.2188	3.15	2.91	4.33	1.72	27000	-0.26

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TABLE I. DASH NUMBERS AND DIMENSIONS -Continued

DASH NUMBER	PART NUMBER 1/ CONE-CUP	d BORE	D OUTSIDE DIA	T BEARING WIDTH	T1 STANDOUT	B CONE WIDTH	C CUP WIDTH	R 2/ MAX SHAFT FILLET RADIUS	D1 FLANGE DIA	C1 FLANGE WIDTH	db BACKING SHOULDER DIAMETERS SHAFT	da BACKING SHOULDER DIAMETERS SHAFT	Da BACKING SHOULDER DIAMETERS HOUSING	K FACTOR	BASIC DYNAMIC LOAD RATINGS (LB) RADIAL	a 3/ EFFECTIVE LOAD CENTER
081 *	641-632-B	2.6250	5.3750	1.6250	0.6560	1.6250	1.2500	0.14	5.6520	0.2810	3.27	3.03	4.92	1.61	37400	-0.44
082	399A-394A-B	2.6875	4.3307	0.8661	0.3120	0.8660	0.7411	0.09	4.5147	0.1870	3.07	2.91	4.17	1.45	16200	-0.03
083	560-S-552-B	2.6875	4.8750	1.5000	0.5625	1.1440	1.1875	0.14	5.1210	0.2500	3.27	2.99	4.57	1.69	30700	-0.37
084	H715343-H715310-B	2.6875	5.5000	1.8125	0.6875	1.8125	1.4375	0.14	6.0000	0.3125	3.54	3.31	5.31	1.24	43000	-0.34
085	29675-29620-B	2.7500	4.4375	1.0000	0.4063	1.0000	0.7500	0.06	4.5900	0.1563	3.15	3.03	4.33	1.20	18000	-0.04
086	LM613449-LM613410-B	2.7500	4.4375	0.8750	0.4375	0.8660	0.6250	0.06	4.6875	0.1875	3.07	2.99	4.33	1.40	16400	-0.00
087	482-472-B	2.7500	4.7244	1.1730	0.4368	1.1420	0.9542	0.14	4.9384	0.2180	3.27	3.03	4.53	1.52	23100	-0.16
088	H414249-H414210-B	2.8125	5.3750	1.6250	0.6550	1.6250	1.2500	0.14	5.6550	0.2800	3.50	3.27	5.12	1.62	41800	-0.43
089	644-632-B	2.8125	5.3750	1.6250	0.6560	1.6250	1.2500	0.14	5.6520	0.2810	3.43	3.19	4.92	1.61	37400	-0.44
090	33287-33462-B	2.8750	4.6250	1.1875	0.4375	1.1875	0.9375	0.14	4.8084	0.1875	3.43	3.15	4.49	1.34	22300	-0.11
091 *	567-563-B	2.8750	5.0000	1.4375	0.5625	1.4240	1.1250	0.14	5.2460	0.2500	3.46	3.19	4.76	1.69	30700	-0.32
092	42688-42620-B	3.0000	5.0000	1.1875	0.5313	1.2205	0.8750	0.25	5.2460	0.2188	3.78	3.31	4.88	1.39	26100	-0.11
094	6461A-6420-B	3.0000	5.8750	2.1250	0.6875	2.1350	1.7500	0.38	6.1835	0.3125	4.25	3.52	5.51	1.61	55100	-0.59
095	LM814849-LM814810-B	3.0625	4.6250	1.0000	0.4375	1.0000	0.7500	0.14	4.8084	0.1875	3.58	3.35	4.57	1.15	19200	-0.09
095	34306-34478-B	3.0625	4.7812	0.9688	0.4688	0.9060	0.6875	0.14	4.9648	0.1875	3.54	3.31	4.65	1.15	19200	-0.06
097	42690-42620-B	3.0625	5.0000	1.1875	0.5313	1.2205	0.8750	0.14	5.2460	0.2188	3.58	3.35	4.88	1.39	26100	-0.11
098	H715348-H715310-B	3.0625	5.5000	1.8125	0.6875	1.8125	1.4375	0.14	6.0000	0.3125	3.86	3.48	5.31	1.24	43000	-0.34
099	L116149-L116110-B	3.2500	4.5625	0.8125	0.3125	0.8438	0.6563	0.06	4.7187	0.1563	3.54	3.46	4.45	1.90	8700	-0.05
100	47686-47620-B	3.2500	5.2500	1.3125	0.5000	1.3125	1.0313	0.14	5.4650	0.2188	3.82	3.54	5.12	1.44	29100	-0.17

* INACTIVE FOR NEW DESIGN AFTER 20 MAR 1978.

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3/ MINUS VALUE INDICATES LOAD CENTER INSIDE CONE BACKFACE.

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TABLE I. DASH NUMBERS AND DIMENSIONS - Continued

DASH NUMBER	PART NUMBER 1/ CONE-CUP	d BORE	D OUTSIDE DIA	T BEARING WIDTH	T1 STANDOUT	B CONE WIDTH	C CUP WIDTH	R 2/ MAX SHAFT FILLET RADIUS	D1 FLANGE DIA	C1 FLANGE WIDTH	db BACKING SHOULDER DIAMETERS SHAFT	da BACKING SHOULDER DIAMETERS SHAFT	Da BACKING SHOULDER DIAMETERS HOUSING	K FACTOR	BASIC DYNAMIC LOAD RATINGS (LB) RADIAL	a 3/ EFFECTIVE LOAD CENTER
101	580-572-B	3.2500	5.5115	1.4375	0.5625	1.4212	1.1250	0.14	5.7575	0.2500	3.86	3.58	5.28	1.45	29800	-0.21
102	663-652-B	3.2500	6.0000	1.6250	0.6562	1.6250	1.2500	0.14	6.2772	0.2812	3.90	3.62	5.55	1.43	39600	-0.31
103	497-493-B	3.3750	5.3750	1.1875	0.5305	1.1720	0.8750	0.14	5.5890	0.2180	3.90	3.66	5.16	1.31	25000	-0.03
104	665-652-B	3.3750	6.0000	1.6250	0.0562	1.6250	1.2500	0.14	6.2772	0.2812	4.02	3.74	5.55	1.43	39600	-0.31
105	841-832-B	3.3750	6.6250	2.1250	0.8750	2.2190	1.6250	0.14	6.9960	0.3750	4.09	3.82	6.10	1.95	65100	-0.73
106	759-752-B	3.5000	6.3750	1.8750	0.6870	1.9000	1.5000	0.14	6.6830	0.3120	4.17	3.90	5.91	1.71	52300	-0.47
107	6580-6535-B	3.5000	6.3750	2.1250	0.7500	2.1693	1.6875	0.14	6.7500	0.3125	4.29	4.01	6.10	1.46	51900	-0.52
108	850-832-B	3.5000	6.6250	2.1250	0.8750	2.2190	1.6250	0.14	6.9960	0.3750	4.17	3.94	6.10	1.95	65100	-0.73
109	760-752-B	3.5625	6.3750	1.8750	0.6870	1.9000	1.5000	0.14	6.6830	0.3120	4.21	3.98	5.91	1.71	52300	-0.47
110	47890-47825-B	3.6250	5.6250	1.3125	0.5937	1.3750	1.0313	0.14	5.8710	0.3125	4.21	3.98	5.59	1.30	31800	-0.04
111	47896-47825-B	3.7500	5.6250	1.3125	0.5937	1.3750	1.0313	0.14	5.8710	0.3125	4.33	4.06	5.59	1.30	31800	-0.04
112	594-592-B	3.7500	6.0000	1.5625	0.6250	1.4300	1.1875	0.14	6.2460	0.2500	4.33	4.09	5.67	1.32	34800	-0.10
113	594A-592-B	3.7500	6.0000	1.5625	0.6250	1.4300	1.1875	0.20	6.2460	0.2500	4.45	4.09	5.67	1.32	34800	-0.10
114	77375-77675-B	3.7500	6.7500	1.8750	0.6875	1.9000	1.5000	0.14	7.0584	0.3125	4.45	4.17	6.34	1.59	54300	-0.38
115	42381-42587-B	3.8125	5.8750	1.2500	0.5000	1.1406	0.9688	0.14	6.0898	0.2188	4.33	4.09	5.98	1.19	26400	0.12
116	52400-52637-B	4.0000	6.3750	1.4375	0.6562	1.4219	1.0313	0.14	6.6210	0.2500	4.61	4.37	6.10	1.23	36100	-0.01
117	687-672-B	4.0000	6.6250	1.6250	0.7187	1.6250	1.1875	0.14	6.9030	0.2812	4.65	4.41	6.30	1.24	39800	-0.11
118	861-854-B	4.0000	7.5000	2.2500	0.8750	2.2650	1.7500	0.31	7.8710	0.3750	5.08	4.49	6.85	1.74	73200	-0.60
119	HH221449-HH221410-B	4.0000	7.50000	2.2500	0.8750	2.2650	1.8125	0.31	7.8710	0.4375	5.16	4.56	7.05	1.74	85200	-0.59
120	941-932-B	4.0000	8.3750	2.6250	0.9375	2.6250	2.1250	0.28	8.8085	0.4375	5.12	4.61	7.83	1.79	83300	-0.78

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TABLE I. DASH NUMBERS AND DIMENSIONS - Continued

DASH NUMBER	PART NUMBER 1/ CONE-CUP	d BORE	D OUTSIDE DIA	T BEARING WIDTH	T1 STANDOUT	B CONE WIDTH	C CUP WIDTH	R 2/ MAX SHAFT FILLET RADIUS	D1 FLANGE DIA	C1 FLANGE WIDTH	db BACKING SHOULDER DIAMETERS SHAFT	da BACKING SHOULDER DIAMETERS SHAFT	Da BACKING SHOULDER DIAMETERS HOUSING	K FACTOR	BASIC DYNAMIC LOAD RATINGS (LB) RADIAL	a 3/ EFFECTIVE LOAD CENTER
121	782-772-B	4.1250	7.1250	1.8750	0.6875	1.8900	1.5000	0.14	7.4330	0.3125	4.80	4.57	6.61	1.51	51400	-0.32
122	56425-56650-B	4.2500	6.5000	1.4375	0.6250	1.4375	1.0625	0.14	6.7460	0.2500	4.84	4.61	6.38	1.18	36600	0.08
123	64450-64700-B	4.5000	7.0000	0.9062	0.7188	1.6250	1.1875	0.14	7.2772	0.2813	5.16	4.92	6.85	1.13	44300	0.05
124	71450-71750-B	4.5000	7.5000	1.8750	0.8125	1.9375	1.3750	0.14	7.8080	0.3125	5.20	4.92	7.13	1.40	58300	-0.26
125	938-932-B	4.5000	8.3750	2.6250	0.9375	2.6250	2.1250	0.28	8.8085	0.4375	5.55	5.04	7.83	1.79	83300	-0.78
126	68462-68712-B	4.6250	7.1250	1.3750	0.6563	1.2500	1.0000	0.14	7.4022	0.2813	5.20	4.92	6.77	1.18	31800	-0.21
127	L624549-L624510-B	4.7500	6.3125	0.8438	0.3438	0.8438	0.6563	0.06	6.4647	0.1563	5.08	5.00	6.18	1.34	17300	0.33
128	795-792-B	4.7500	8.1250	1.8750	0.8125	1.8750	1.3750	0.13	8.4336	0.3125	5.47	5.28	7.80	1.27	57000	-0.07
129	48290-48220-B	5.0000	7.1875	1.5625	0.5000	1.5000	1.3125	0.14	7.4335	0.2500	5.55	5.31	6.97	1.91	43200	-0.22
130	67388-67322-B	5.0000	7.7500	1.8125	0.5938	1.8125	1.5000	0.14	8.0272	0.2813	5.67	5.43	7.52	1.70	56800	-0.25
131	95500-95925-B	5.0000	9.2500	2.5000	1.0000	2.5000	1.9375	0.25	9.6834	0.4375	6.06	5.59	8.54	1.58	89400	-0.55
132	799A-792-B	5.1250	8.1250	1.8750	0.8125	1.8750	1.3750	0.14	8.4336	0.3125	5.83	5.59	7.80	1.27	57000	-0.07
133	L327249-L327210B	5.2500	6.9688	1.0000	0.3437	1.0313	0.8125	0.06	7.1210	0.1562	5.59	5.51	6.81	1.68	24000	0.16
134	67390-67322-B	5.2500	7.7500	1.8125	0.5938	1.8125	1.5000	0.14	8.0272	0.2813	5.87	5.63	7.52	1.70	56800	-0.25
135	95528-95925-B	5.2500	9.2500	2.5000	1.0000	2.5000	1.9375	0.19	9.6834	0.4375	6.18	5.83	8.54	1.58	89400	-0.55
136	48393-48320-B	5.3750	7.5000	1.5625	0.4688	1.5625	1.3125	0.14	7.7148	0.2188	5.94	5.67	7.32	1.82	43800	-0.16
137	74550-74850-B	5.5000	8.5000	1.8750	0.8125	1.8750	1.3750	0.14	8.8084	0.3125	6.22	5.94	8.23	1.20	57800	0.09
138	898-892-B	5.5000	9.0000	2.2500	0.8750	2.2500	1.7500	0.14	9.3710	0.3750	6.30	6.02	8.50	1.39	83300	-0.24

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TABLE I. DASH NUMBERS AND DIMENSIONS

DASH NUMBER	PART NUMBER 1/ CONE-CUP	d BORE	D OUTSIDE DIA	T BEARING WIDTH	T1 STANDOUT	B CONE WIDTH	C CUP WIDTH	R 2/ MAX SHAFT FILLET RADIUS	D1 FLANGE DIA	C1 FLANGE WIDTH	db BACKING SHOULDER DIAMETERS SHAFT	da BACKING SHOULDER DIAMETERS SHAFT	Da BACKING SHOULDER DIAMETERS HOUSING	K FACTOR	BASIC DYNAMIC LOAD RATINGS (LB) RADIAL	a 3/ EFFECTIVE LOAD CENTER
139	48685-48620-B	5.6250	7.8750	1.6250	0.5000	1.5625	1.3437	0.14	8.0898	0.2187	6.22	5.94	7.64	1.74	43500	-0.12
140	36690-36620-B	5.7500	7.6250	1.1250	0.3750	1.1250	0.9063	0.06	7.7772	0.1563	6.10	6.02	7.48	1.59	32000	0.19
141	82576-82950-B	5.7500	9.5000	2.2500	0.8750	2.2300	1.7500	0.14	9.8710	0.3750	6.54	6.30	8.90	1.32	82500	-0.14
142	HM231149-HM231115-B	5.8750	9.5000	2.2500	0.8750	2.2300	1.7500	0.14	9.8710	0.3750	6.57	6.42	8.82	1.83	86500	-0.45
143	L630349-L630310-B	6.0000	7.5625	0.9843	0.3925	0.9449	0.7480	0.08	7.7705	0.1562	6.38	6.22	7.44	1.40	13900	0.40
144	99600-99100-B	6.0000	10.0000	2.6250	1.1875	2.6250	1.8750	0.28	10.4320	0.4375	7.13	6.68	9.37	1.43	91400	-0.48

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REQUIREMENTS:

1. Material: Cones, cups, and rollers: bearing quality - carburizing grade alloy or through-hardening grade alloy steel in accordance with ASTM-A295, ASTM-A534, or ASTM-A535. Cage: carbon steel (one piece stamped).

2. Hardness: Cones, cups and rollers: 58 to 64HRC.

3. Tolerances: Standard grade (AFBMA Class 4) (allowable tolerances are in 0.0001 inch).

Cone bore (d)			
Size range		Tolerance	
Over	Incl.	Plus	Over
0 3.0000	3.0000 6.0000	5 10	0 0

Cup diameter (D)			
Size range		Tolerance	
Incl.	Plus	Minus	
0.0000 12.0000	12.0000 24.0000	10 20	0 0

Standout (T1)			
Bore size range		Tolerance	
Over	Incl.	Plus	Minus
0 4.0000	4.0000 6.0000	80 140	0 100

Cup flange OD (D1)			
Size range		Tolerance	
Over	Incl.	Plus	Minus
0	12.0000	20	0

Assembled bearing maximum radial runout		
Cup O.D. (D)		Tolerance
Over	Incl.	
0	24.0	20

4. Effective load center: Dimension (o) locates a point on the cone axis which is the center of pressure of all resisting forces set up by the rollers. All moments should be calculated from this point when determining bearing loading and shaft stresses. A plus value of (a) indicates that the center is outside the cone backface.

5. Operating temperature: Recommended operating temperature not to exceed 230″ F.

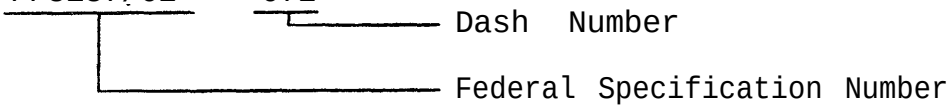
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6. Lubrication: Bearings shall be furnished without lubrication. All bearings shall be preserved in accordance with MIL-P-197.

7. Normal angle: A normal angle bearing has a contact angle (a) between 10 and 19 degrees. The contact angle is the single between the line of the roller load and a plane perpendicular to the bearing axis.

8. Port number: The part number consists of the letters FFB, the basic federal specification number, the specification shed number and a 'dash number from table I.

EXAMPLE: FF8187/02 - 072



9. For design feature purposes, this standard takes precedence over procurement documents referenced herein.

10* Referenced documents shall be of the issue in effect on the date of invitation for bids, or request for proposal.

Note: Inch units can be converted to soft metric units as follows:

x	25.4	=	Millimeters

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INTERCHANGEABILITY TABLE

MILITARY STD PART NUMBERS	SUBSTITUTIVE PART NUMBERS	MILITARY STD PART NUMBERS	SUBSTITUTIVE PART NUMBERS
MS19082-1	FFB187/02-005	MS19082-26	FFB187/02-117
MS19082-2	FFB187/02-012	MS19082-27	FFB187/02-122
MS19082-3	FFB187/02-020	MS19082-28	FFB187/02-140
MS19082-4	FFB187/02-032	-	-
MS19082-5	FFB187/02-041	-	-
MS19082-6	FFB187/02-055	-	-
MS19082-7	FFB187/02-065	-	-
MS19082-8	FFB187/02-074	-	-
MS19082-9	FFB187/02-093	-	-
MS19082-10	FFB187/02-118	-	-
MS19082-11	FFB187/02-001	-	-
MS19082-12	FFB187/02-002	-	-
MS19082-13	FFB187/02-011	-	-
MS19082-14	FFB187/02-019	-	-
MS19082-15	FFB187/02-021	-	-
MS19082-16	FFB187/02-023	-	-
MS19082-17	FFB187/02-024	-	-
MS19082-18	FFB187/02-029	-	-
MS19082-19	FFB187/02-034	-	-
MS19082-20	FFB187/02-052	-	-
MS19082-21	FFB187/02-054	-	-
MS19082-22	FFB187/02-073	-	-
MS19082-23	FFB187/02-080	-	-
MS19082-24	FFB187/02-082	-	-
MS19082-25	FFB187/02-112		

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